More on fixed point theorems

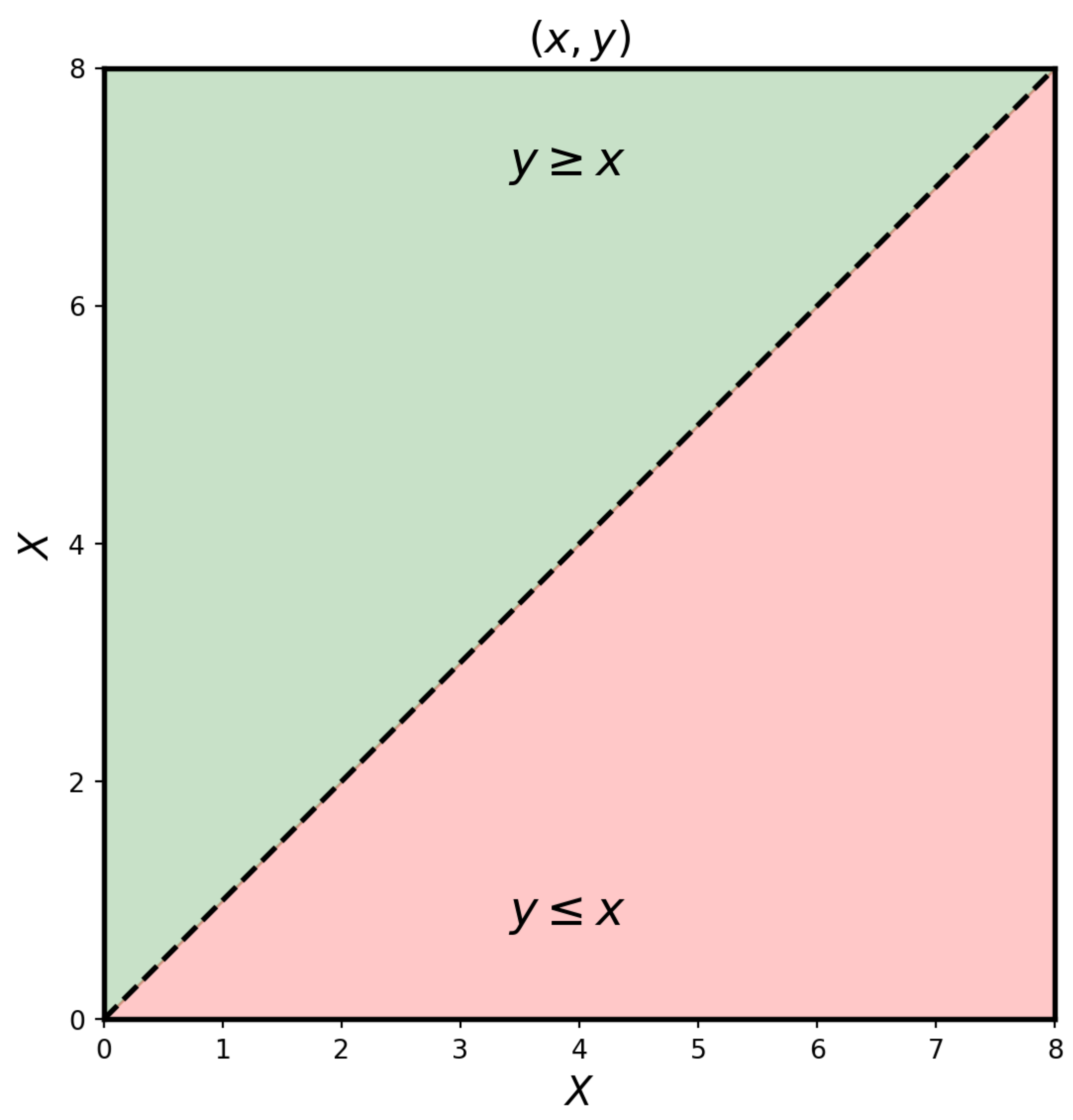
Tarski FPT

A fixed point theorem that is not based on continuity.

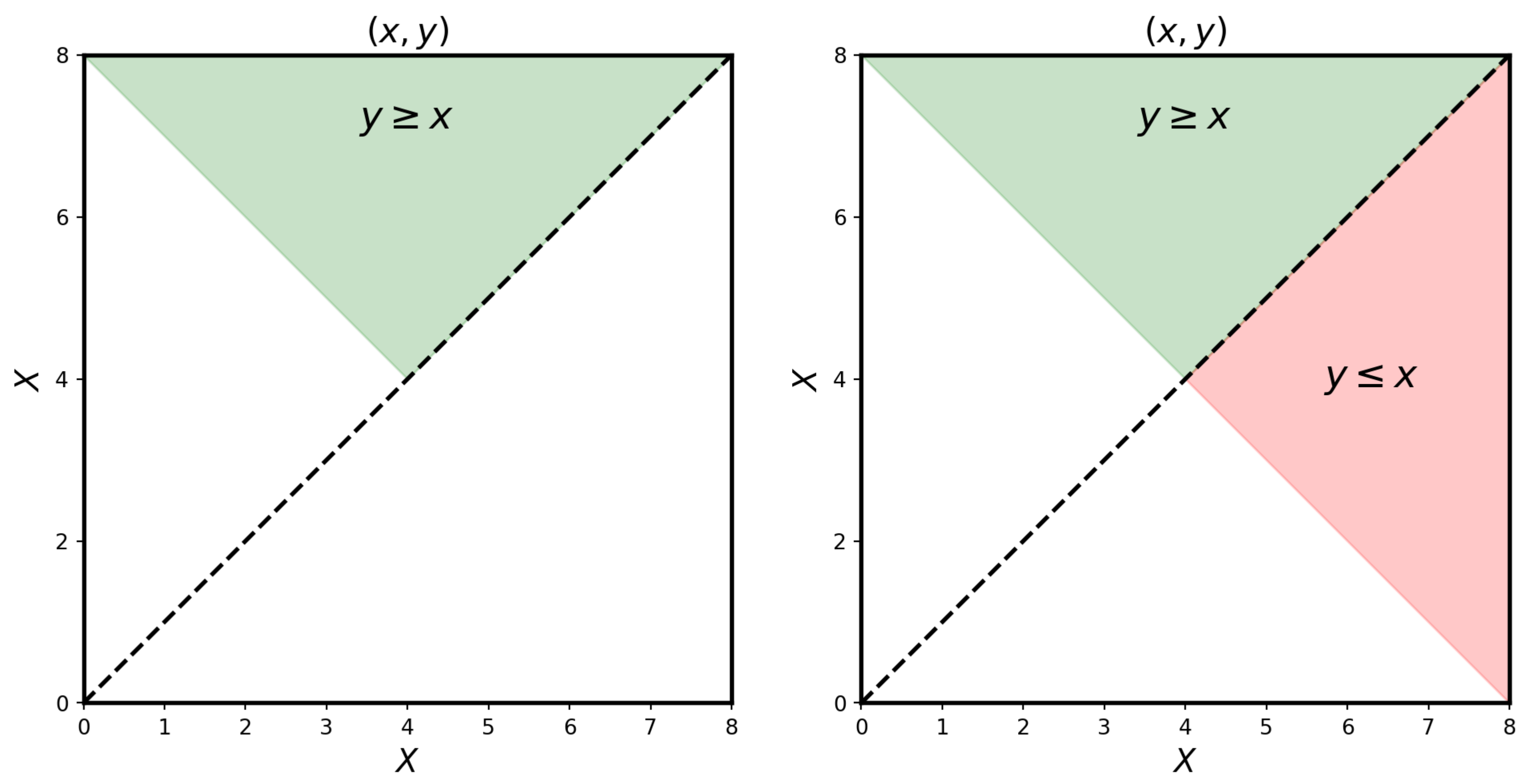
Preliminaries

- A binary relation R on a set X is a set $R \subseteq X \times X$.
  $x\,R\,y$ means $(x, y) \in R$.
- A partial order, denoted by $\geq$, is a binary relation that is reflexive, transitive, and antisymmetric, that is
  $(\forall x, y \in X)$, $[x \geq y \;\wedge\; y \geq x] \implies x = y$.
- A set X with a partial order $\geq$ is a partially ordered set, denoted $(X, \leq)$.
- Henceforth, fix $(X, \geq)$.
  - $(\forall x, y \in X)$ [$x > y$ means $x \geq y$ and $x \neq y$].
  - $(\forall A \subseteq X)$ $(A, \geq)$ is the partially ordered set where $\geq$ is the restriction of $\geq$ to A.

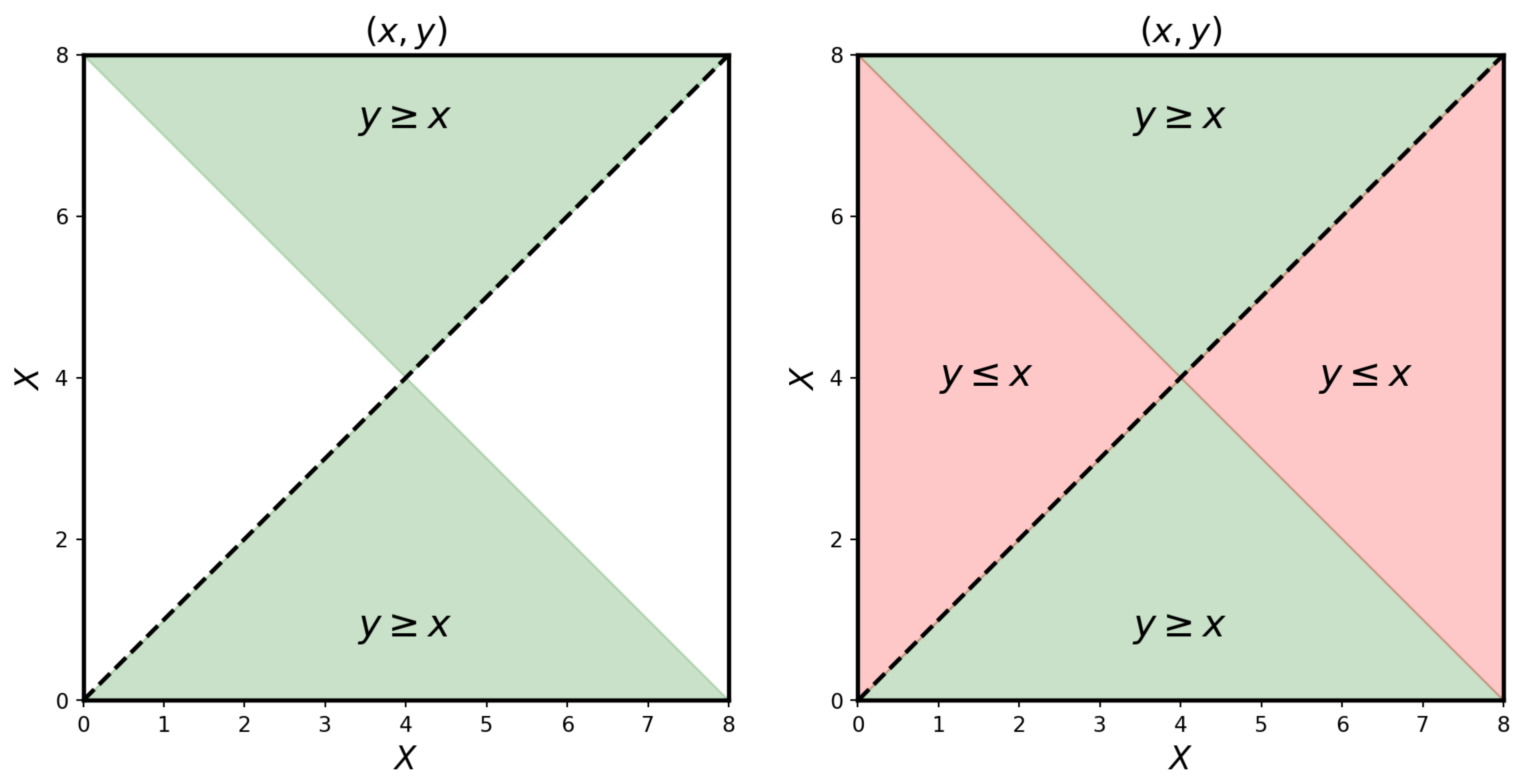
Standard order



Partial order, not complete



Partial order, complete



- A binary relation R is total if

$$x \neq y \implies [(x\ R\ y) \vee (y\ R\ x)].$$

A total binary relation is ``almost” complete; the only possibly missing pairs are some (x, x).

- $C \subseteq X$ is a chain in $(X, \geq)$ if $(C, \geq)$ is total, i.e.

$$(\forall x, y \in X)\ (x \geq y) \vee (y \geq x).$$

Upper bound, maximal, greatest

- Let $A \subseteq X$.
  - $x \in X$ is an upper bound for A if $(\forall y \in A)\ x \geq y$.
  - $x \in A$ is maximal in A if $(\nexists y \in A)\ y \geq x$.
  - $x \in A$ is a greatest element of A if $(\forall y \in A)\ x \geq y$.
- A nonempty subset of X has at most one greatest element. If a greatest element exists, it is maximal.
- Lower bound, minimal element, and least element, are defined analogously.

Supremum, infimum

- $x \in X$ is the supremum of A, denoted $x = \bigwedge A$ if it is its least upper bound.
- The Infimum of A is defined analogously and denoted $x = \bigvee A$.
- Suprema and infima, often abreviated sup and inf, need not exist.
- $x \vee y$ is the sup and $x \wedge y$ is the inf of $\{x, y\}$.
- For linear orders $x \vee y = \max\{x, y\}$ and $x \wedge y = \min\{x, y\}$.

Zorn's lemma and the axiom of choice

Definition 1 (Axiom of choice) If $\{A_i\}_{i \in I}$ is a collection of non-empty sets indexed by $i \in I$ where I is non-empty. Then there is a function $f : I \to \bigcup_{i \in I} A_i$ such that $(\forall i \in I)\ f(i) \in A_i$.

Lemma 1 (Zorn) If every chain in a partially ordered set X has an upper bound, then X has a maximal element.

- The axiom of choice and Zorn's lemma are equivalent.

Monotone functions

- Let $(X, \geq_X)$ and $(Y, \geq_Y)$ be two partially ordered sets.
- A function $f : X \to Y$ is monotone if $x \geq_X z \implies f(x) \geq_Y f(z)$.
- f is strictly monotone if $x >_X z \implies f(x) >_Y f(y)$.
- Monotone functions are also called increasing or nondecreasing functions.
- Decreasing or nonincreasing functions and strictly decreasing functions are defined analogously.

Knaster-Tarski

See (Knaster 1928), (Tarski 1955), and (Aliprantis and Border 2006).

Theorem 1 (Knaster–Tarski Fixed Point Theorem) Let $(X, \geq)$ be a partially ordered set with the property that every chain in X has a supremum. Let $f : X \to X$ be monotone, and assume that there exists $a \in X$ with $a \leq f(a)$. Then the set of fixed points of f is nonempty and has a maximal fixed point.

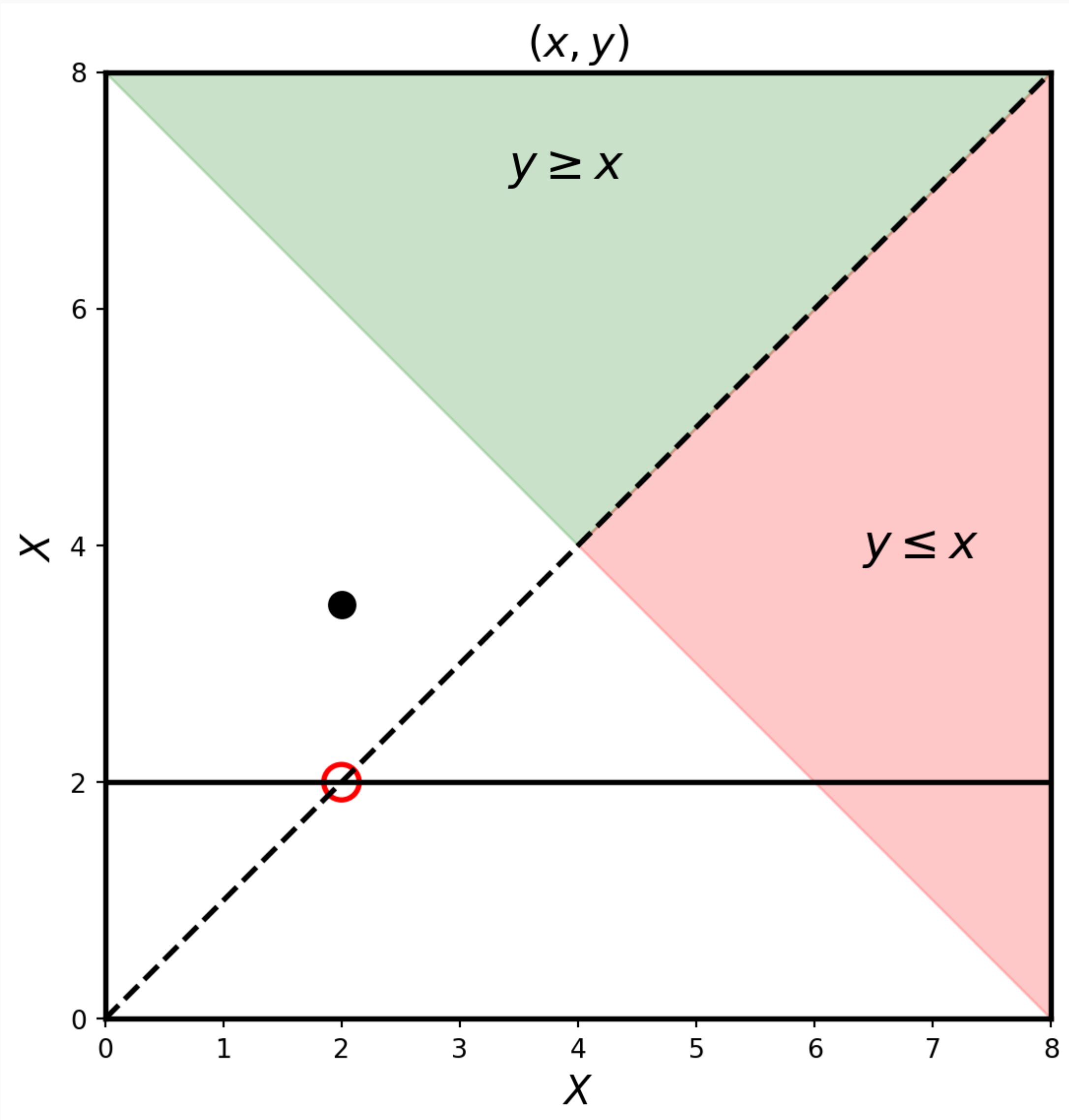
Proof

- Let $P = \{x \in X : x \leq f(x)\}$.
- $(P, \geq)$ is partially ordered, $a \in P$.
- Let C be a chain in P, and $b = \bigwedge P$. $b \in X$.
- $(\forall c \in C)\ c \leq b$. Then $f(c) \leq f(b)$.
- $(\forall c \in C)\ c \leq f(c)$ because $c \in C \subseteq P$.
- It follows that $f(b)$ is an upper bound for C.
- Since b is the least such upper bound, $b \leq f(b)$. Therefore, $b \in P$.
- Thus the supremum of any chain in P belongs to P.

- Zorn's Lemma implies P has a maximal element, say x'.
- Now $x' \leq f(x')$, since $x' \in P$.
- Since f is monotone, $f(x') \leq f(f(x'))$.
- This means that $f(x')$ belongs to P.
- Since x' is a maximal element of P, $x' = f(x')$.
- Furthermore, if x is a fixed point of f, then $x \in P$.
- This shows that x' is a maximal fixed point of f.

QED

Why is $[(\exists a)\ a \leq f(a)]$ necessary?



Related result

- Tarski (1955) has a related fixed-point theorem.
- It strengthens the hypotheses to require $(X, \geq)$ to be a complete lattice, and draws the stronger conclusion that the set of fixed points is also a complete lattice.

Lattice

- A lattice is a partially ordered set in which every pair of elements has a supremum and an infimum.
- It follows by induction that every finite set in a lattice has a supremum and an infimum.
- S is a sublattice of a lattice L if $S \subseteq L$ and

$$(\forall a, b \in S)\ [(a \wedge b \in S) \wedge (a \vee b \in S)].$$

- A complete lattice is a lattice in which every nonempty subset A has a sup $\bigvee A$ and an inf $\bigwedge A$.
- The infimum of a complete lattice is denoted 0, and the supremum is denoted 1.

Tarski fixed-point theorem

Theorem 2 (Tarski) If $(X, \geq)$ is a complete lattice, and $f : X \to X$ is monotone, then the set F of fixed points of f is nonempty and $(F, \geq)$, is itself a complete lattice.

- The theorem has been extended to cover increasing correspondences. See Smithson (1971), Vives (1990), and Zhou (1994); Echenique (2005) provides simple, constructive proofs.

Interpretation

- F is a complete lattice but may not be a sublattice of X.
- Example. $X = \{0, 1, a, b, b'\}$ with the partial order $\geq$ defined by $1 \geq a \geq b \geq 0$ and $1 \geq a \geq b' \geq 0$ (and all comparisons implied by transitivity and reflexivity).
- b and b' are not comparable.
- Define $f : X \to X$, $f(x) = x$ for $x \neq a$ and $f(a) = 1$. f is monotone.
- $F = \{0, b, b', 1\}$ is a complete lattice.
- Let $B = \{b, b'\}$. $\bigvee B = 1$ if B in lattice F; $\bigvee B = a$ if B in lattice X.

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